

Design and Development of a Geolocation-Based Intelligent System for Reporting, Monitoring, and Analyzing Electricity Outages in Cameroon

Level 2 Software Engineering Project Proposal

1. Project Overview

The proposed project is a digital platform that allows citizens to report electricity outages, automatically capture their geographic location, visualize outage incidents on an interactive map, group nearby reports into potential common incidents, monitor restoration, and generate historical analytics.

2. Problem Statement

Electricity outages are a recurring challenge in many parts of Cameroon. When power goes off, users often have limited information about whether the outage is local or widespread, whether it has already been reported, how many people may be affected, or whether electricity has been restored. Information is frequently fragmented across phone calls, messaging groups, social media, and informal communication.

The proposed system centralizes citizen-generated reports and transforms them into structured, geolocated information that can be monitored and analyzed.

3. Main Objective

To design and develop a geolocation-based intelligent system capable of collecting, aggregating, monitoring, and analyzing electricity outage reports in urban areas of Cameroon.

4. Specific Objectives

- Allow users to report electricity outages from a mobile application.
- Automatically capture the geographic position, date, and time of each report.
- Display reported incidents on an interactive map.
- Group geographically and temporally related reports into potential common outage events.
- Provide outage status management: reported, active/investigating, and restored.
- Notify users about relevant nearby outage events and restoration updates.
- Provide an administrative dashboard with statistics and historical analytics.
- Create a foundation for future predictive analysis based on historical outage data.

5. Target Users

Citizens: report outages and receive information about nearby incidents.

Administrators: validate, monitor, manage, and analyze reported incidents.

Service operators: potentially receive structured incident information through future partnerships.

Businesses and institutions: monitor electricity reliability at their locations.

6. Core Functionalities

Module	Main Functions
Authentication	Registration, login, user profile, role management
Outage Reporting	Report outage, type, description, GPS location, timestamp
Geolocation	Automatic position capture and neighborhood identification
Interactive Map	Display active, recent, and restored incidents
Intelligent Detection	Group nearby reports into potential common outage events
Severity	Estimate incident severity from report volume and geographic coverage
Notifications	Nearby outage alerts and restoration notifications
History	Past incidents, duration, location, and report count
Administration	Incident validation, status management, user management
Analytics	Outages by area, duration, frequency, peak periods, trends

7. Intelligent Component

The intelligence of the system can begin with a rule-based geographic and temporal clustering mechanism. For example, if several users report outages within a defined geographic radius and within a short time window, the system can treat them as evidence of a common outage event rather than unrelated reports.

Example: Reports from Bastos at 18:42, 18:44, 18:46 and 18:48 can be grouped into a single potential outage event. The system can then calculate report count, approximate affected area, severity, and duration.

A future version can introduce machine-learning models after enough historical data has been collected. Predictive analytics should remain an optional advanced feature for the Level 2 prototype.

8. Suggested Technical Architecture

Mobile Application → REST API → Authentication & Business Logic → PostgreSQL/PostGIS Database → Notification Service + Mapping Service → Administrative Web Dashboard

9. Recommended Technologies

Layer	Recommended Technology
Mobile	Flutter
Backend	Node.js + Express (or Laravel)
Database	PostgreSQL + PostGIS
Maps	OpenStreetMap + Leaflet
Notifications	Firebase Cloud Messaging
Authentication	JWT or Firebase Authentication
Dashboard	React
Version Control	Git + GitHub/GitLab

10. Main Database Entities

Users: id, name, email, phone, password, role, location, creation date.

Outage Reports: id, user, latitude, longitude, neighborhood, type, description, reported time, status.

Outage Events: id, area, coordinates, severity, start time, end time, status, report count.

Notifications: id, user, title, message, type, read status, creation date.

11. Business Model and Profitability

The owner should not depend exclusively on citizens paying to report outages. A stronger model combines a free citizen service with premium business and analytics services.

Revenue Stream	Description
Freemium	Free outage reporting and basic information; optional premium monitoring and advanced history.
Business Subscriptions	Companies monitor offices, stores, branches, warehouses, or other locations.
Reliability Reports	Paid electricity-reliability reports for businesses and real-estate users.
Analytics	Aggregated and anonymized infrastructure analytics for organizations and researchers.
Partnerships	Potential integrations and service agreements with relevant institutions.
Advertising	Carefully selected ads for backup-power and energy-related products; secondary revenue stream.

12. Example of a Business Intelligence Product

Electricity Reliability Report — Bastos, Yaoundé

Average monthly outages: 8

Average outage duration: 1h 47min

Longest recorded outage: 5h 21min

Reliability score: 86/100

Such reports could become valuable to hotels, restaurants, offices, supermarkets, factories, schools, and real-estate companies. Any commercial use of data should rely on aggregated/anonymized information and respect applicable privacy requirements.

13. Level 2 MVP Scope

- Authentication and user profiles
- Electricity outage reporting
- GPS location capture
- Interactive outage map
- Basic geographic/temporal incident clustering
- Outage status management
- Push notifications
- Administrative dashboard
- Historical statistics and charts

14. Suggested Defense Demonstration

During the defense, simulate several users reporting outages in nearby locations. Demonstrate how the system receives the reports, groups them into a potential common outage, updates severity, displays the affected area on the map, notifies nearby users, and finally changes the event to RESTORED after several restoration confirmations. Finish by showing the automatically updated dashboard statistics.

15. Academic Positioning

Research Question: How can a geolocation-based software system aggregate citizen-generated reports to detect, monitor, and analyze electricity outages in urban areas of Cameroon?

Core Contribution: Citizen reports → Geographic analysis → Outage detection → Monitoring → Analytics → Decision support.

16. Final Recommendation

For a Level 2 Software Engineering project, the recommended approach is to build a strong MVP rather than attempting a fully predictive AI system from the beginning. The intelligent clustering, geolocation, notifications, database, API, and dashboard are already sufficient to demonstrate substantial Software Engineering skills. Predictive machine learning can be presented as future work.